

DSM TUT QUIZ 3

1. Given the Boolean function

$$F(W, X, Y, Z) = \sum m(1, 3, 7, 11) + d(0, 2, 5)$$

what is the minimal Sum-of-Products (SOP) expression?

2. Simplify the following Boolean functions and design them using universal gates only .

(a) $F(A, B, C, D) = \prod M(1, 3, 5, 7, 13, 15)$

(b) $F(A, B, C, D) = \sum m(1, 3, 6, 9, 11, 12, 14)$

3. Implement the following four Boolean expressions using three half adders:

$$D = A \oplus B \oplus C$$

$$E = ABC + \overline{A}\overline{B}\overline{C}$$

$$F = ABC + (A + B)C$$

$$G = ABC$$

4. Design a 3-bit adder circuit using a ripple carry adder and a CLA (Carry Look-Ahead) adder. What are the trade-offs between these two methods?

5. Write the truth table for the majority detector circuit and design it using basic gates. How can we design the same circuit using a full adder?

6. Write the truth table for a full subtractor and design a 3-bit subtractor using three full subtractors.

7. Design a combinational circuit comprising all of these logics:

$$F_1 = AB + A'C + BC$$

$$T_1 = ABC$$

$$T_2 = A + B + C$$

$$F_2 = F_1T_2 + T_1$$

8. Design a combinational circuit comprising all of these logics:

$$F_1 = AB + A'B'$$

$$F_2 = BC$$

$$F_3 = F_1' + F_2$$

$$F_4 = F_1' \oplus C$$

9. Write the truth table for even parity and odd parity for 3 bits. Design the 3-bit even parity generator and the 3-bit odd parity generator.

10. Design a combinational circuit with four inputs A, B, C, D that outputs Logic High when the number of 1's among the inputs is even, and outputs Logic Low when the number of 1's among the inputs is odd. What is this circuit equivalent to ?